

Supplementary Material

Three Equivalent Routes to Estimating Power Spectra: Averaging, Multitaper, and Smoothing

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Figure, table and equation numbers without an “S” refer to the main text, whose notation (Section 2.1 there) is used throughout: R is the design resolution, $\alpha = N\Delta R/2$ the time–bandwidth product, L the number of tapers, and widths are in units of $1/(N\Delta)$.

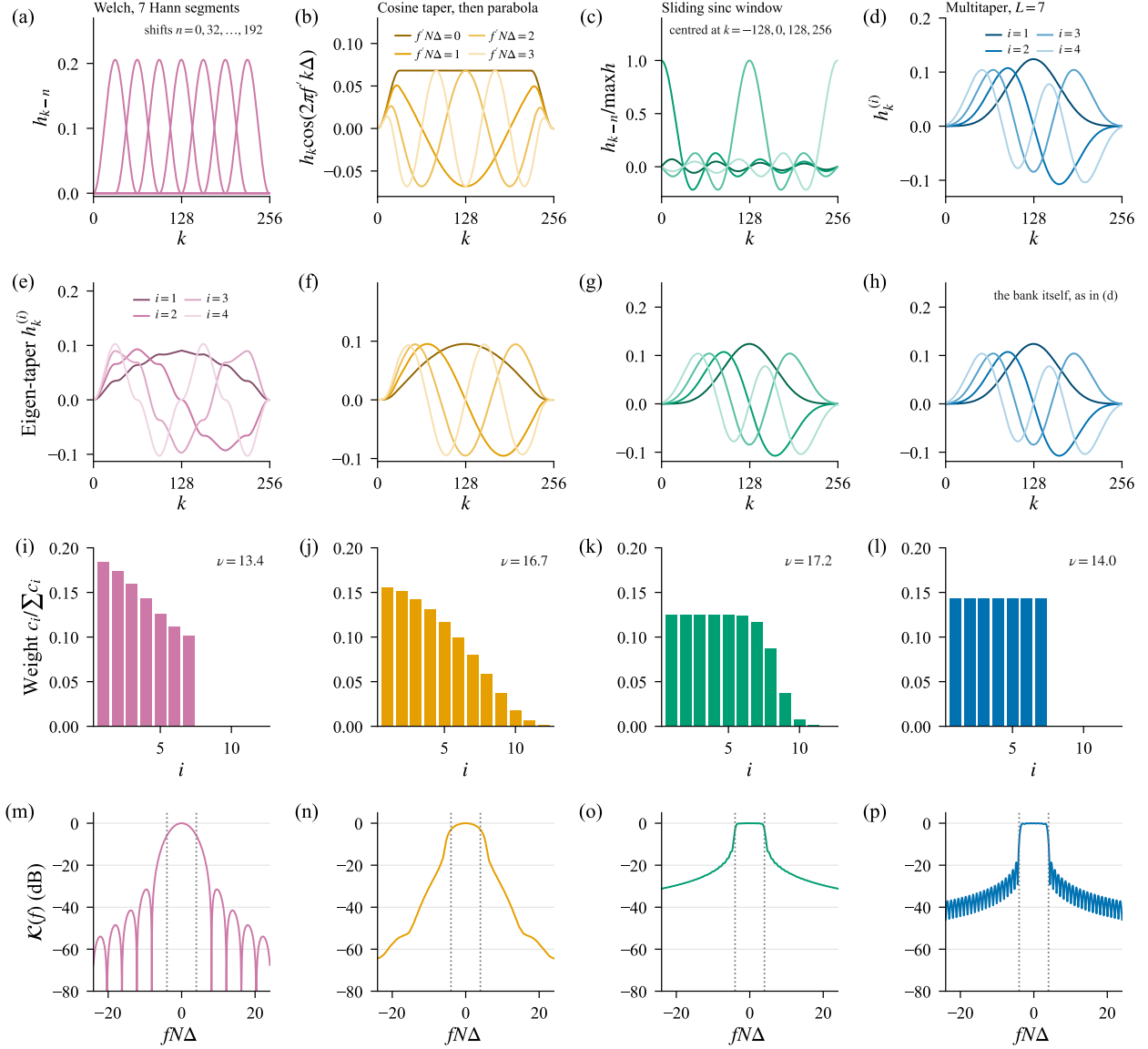


Figure S1. Fig. 5 in full. Columns and colours as in Fig. 5; rows: the bank as constructed (a–d), the first four eigen-tapers of Q (e–h; for the sliding sinc they are the Slepian sequences, and for Thomson’s estimate, whose weights are equal, the Slepian’s are one of many orthonormal bases of the span), the eigen-weight ladder (i–l), and the kernel in dB relative to its peak (m–p; dotted lines at $|f| = R/2$). The half-power widths of the kernels are 5.84, 8.03, 8.03 and 7.41/($N\Delta$), and 0.05%, 0.13%, 1.39% and 0.22% of their power lies outside $|f| \leq R$.

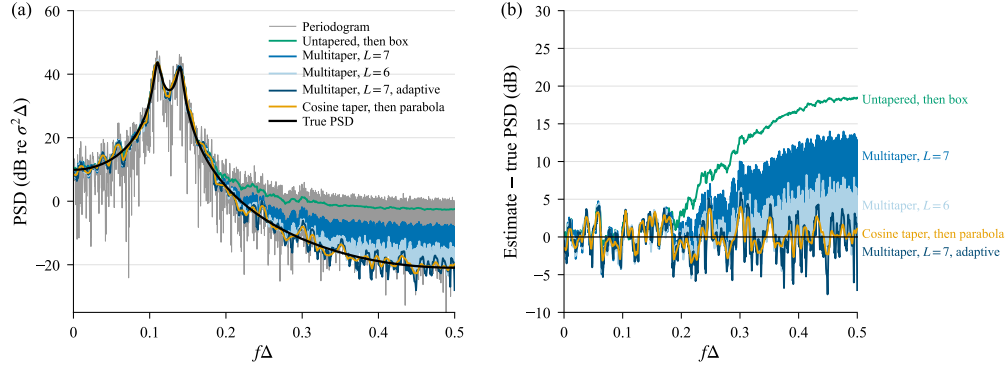


Figure S2. Dropping the leakiest taper. (a) One realization of the AR(4) process ($N = 1024$, $R = 8/(N\Delta)$): the periodogram ($\nu = 2$, grey); the untapered periodogram smoothed with the box ($\nu = 17.2$, green); the equal-weight multitaper estimate with $L = 7$ ($\nu = 14$, blue) and with $L = 6$ ($\nu = 12$, light blue), and the adaptively weighted estimate with $L = 7$ (dark blue; its ν varies with frequency, from 14 where the spectrum is high to 6.5 where it is low); the cosine-tapered periodogram smoothed with a parabola of half-power width R ($\nu = 16.7$, orange); and the true PSD (black). Spectra are in dB relative to $\sigma^2\Delta$, where σ^2 is the variance of the innovations. (b) Each smoothed estimate minus the true PSD, named at its right-hand end. For $f\Delta \geq 0.4$ the untapered box lies 18 dB above the true PSD in the median and the multitaper estimate with $L = 7$ lies 10 dB above it. Dropping the seventh taper lowers that to 4 dB, and the adaptive weights and the smoothed periodogram track the truth.

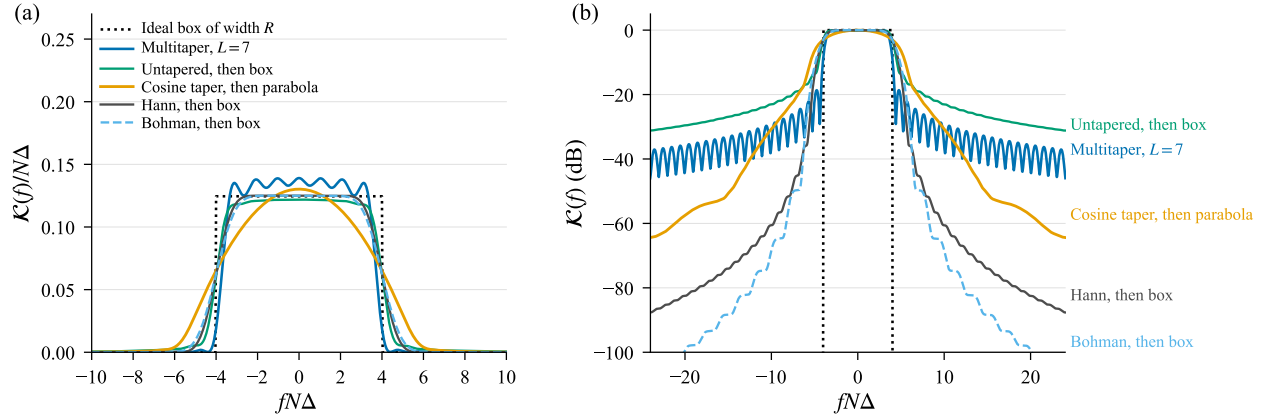


Figure S3. Kernels at the same design resolution ($N = 256$, $R = 8/(N\Delta)$; Table 2), computed exactly, (a) on a linear scale near the main lobe and (b) in dB relative to the peak: the ideal box (dotted); the $L = 7$ multitaper kernel ($\nu = 14.0$, blue); the periodogram after a 25% cosine taper followed by a parabola of the same half-power width ($\nu = 16.7$, orange), which is recipe (b'); and the untapered ($\nu = 17.2$, green), Hann ($\nu = 8.9$, grey) and Bohman ($\nu = 7.3$, light blue, dashed) periodograms followed by the box. In (b) each curve is named at its right-hand end. The half-power widths are $7.4/(N\Delta)$ for the multitaper kernel and $8.0/(N\Delta)$ for the others. At $|f| = 12/(N\Delta)$ the skirts lie between -24 dB (untapered) and -75 dB (Bohman), a range of 51 dB.

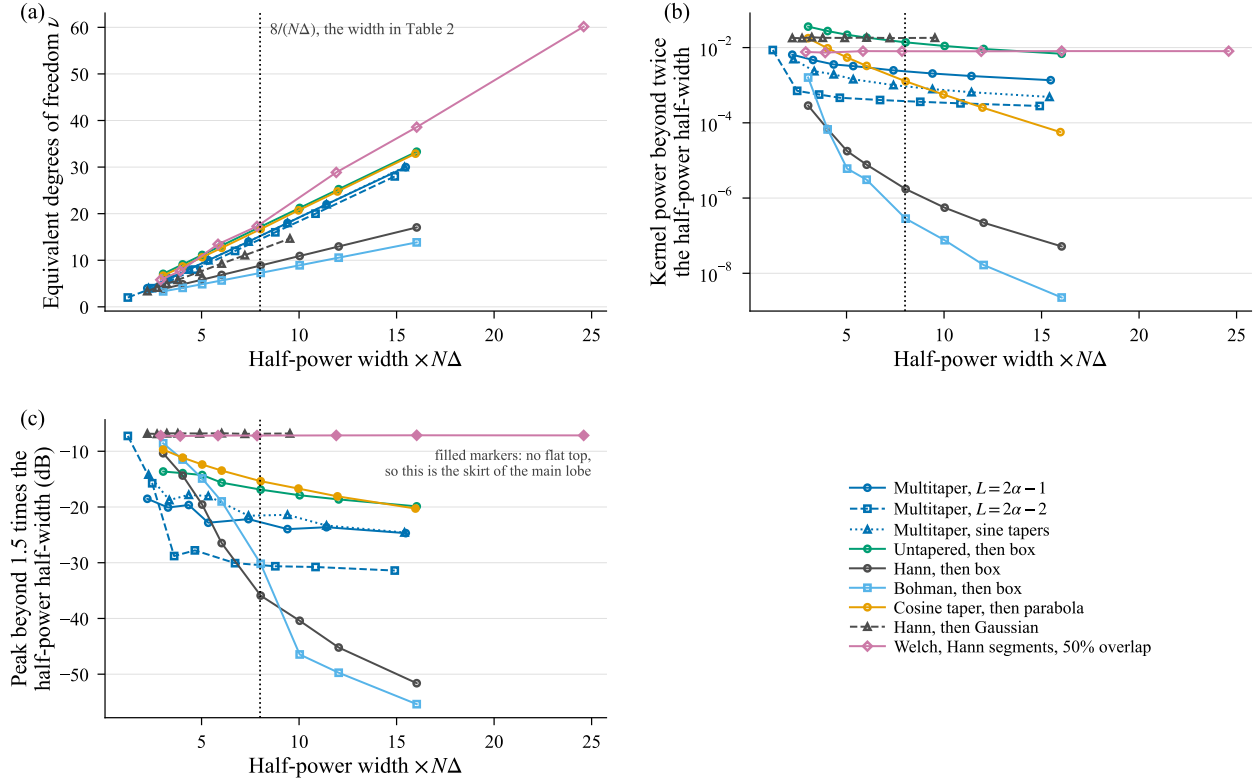


Figure S4. Variance, leakage and side-lobe level against the measured resolution, for $N = 256$. All curves are exact kernel and degrees-of-freedom computations, with no sampling variability. Each method's bandwidth parameter is swept and the quantities are plotted against the resulting half-power width, so methods are compared at equal resolution: multitaper with $L = 2\alpha - 1$ and $2\alpha - 2$ Slepian tapers and with sine tapers; untapered, Hann and Bohman periodograms followed by a box; a cosine-tapered periodogram followed by a parabola; a Hann periodogram followed by a Gaussian; Welch's method with 50% overlap. (a) Equivalent degrees of freedom ν (higher means less variance). (b) Kernel power beyond twice the half-power half-width. (c) The peak of the kernel beyond $1.5\times$ the half-power half-width, in dB relative to its maximum. The Gaussian, parabolic and Welch kernels have no flat top, so for them this quantity is the skirt of the main lobe and not a side lobe, and their markers are filled. The dotted line marks the $8/(N\Delta)$ width of Table 2, and the legend is at the lower right.